

electron/knowledge-graph-runner.cjs

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Overview

`electron/knowledge-graph-runner.cjs` is a new module that builds a deterministic knowledge graph for a codebase. It scans the project, runs a Tree-sitter-based extractor, resolves imports, and writes `knowledge-graph.json` and an optional `diff-overlay.json` into `.novadiff-graph`.

Key changes

- **Imports added** (lines 1-8): `fs`, `fsp`, `path`, `os`, `spawn`, `pathToFileURL`, `createRequire`.
- **Constants** (lines 11-32): `BATCH_SIZE`, `MAX_FILES`, `EXTRACT_TIMEOUT_MS`, `MAX_ANALYZE_LINES`, `SKIP_DIRS`.
- **Repository helpers** (lines 34-43): `resolveRepoRoot`.
- **Node execution helpers** (lines 45-60): `resolveNodeExecutable`, `nodeChildEnv`, `runCommand`.
- **File-system utilities** (lines 147-185): `walkProjectFiles`, `isTextCandidate`.
- **Language & import resolution** (lines 187-279): `detectLanguage`, `fileCategory`, `resolveRelativeImport`.
- **Batch extraction** (lines 281-342): `buildBatchImportData`, `runExtractStructureBatch`.
- **Graph construction helpers** (lines 344-496): `mapStructuralAnalysis`, `buildFileMeta`, `buildLayersFromFiles`, `assignNodeLayers`, `buildDiffOverlay`.
- **Git integration** (lines 497-504): `getGitHash`.
- **Core orchestration** (lines 515-671): `buildKnowledgeGraph` – scans, batches, extracts, assembles the graph, writes files, and reports progress.
- **Exports** (lines 673-677): `resolveRepoRoot`, `ensureGraphCoreBuilt`, `buildKnowledgeGraph`.
- **Build helper** (lines 103-111): `ensureGraphCoreBuilt` compiles the graph engine if needed.

Impact

- Enables deterministic graph generation without LLMs, suitable for CI or diff tooling.
- Limits extraction to 24 files per batch (`BATCH_SIZE`) and 800 files total (`MAX_FILES`) to keep memory usage bounded.
- Requires the `packages/graph-core` package and the `graph/extract-structure.mjs` script; otherwise `ensureGraphCoreBuilt` or `runExtractStructureBatch` will throw.
- Sets `ELECTRON_RUN_AS_NODE` when the Node executable matches the Electron binary, allowing child processes to run as Node.
- Emits progress via the optional `onProgress` callback and records timestamps and the current Git commit hash in the meta file.

Risks & follow-ups

- **Missing core:** `packages/graph-core` must exist and be built; otherwise `buildKnowledgeGraph` fails.
- **Extraction script:** `graph/extract-structure.mjs` must be present; its absence triggers a clear error.
- **Timeouts:** `EXTRACT_TIMEOUT_MS` (12 min) may need adjustment for very large projects; monitor for "Timed out" errors.
- **Path handling:** `resolveRelativeImport` normalizes POSIX paths; verify cross-platform behavior on Windows.